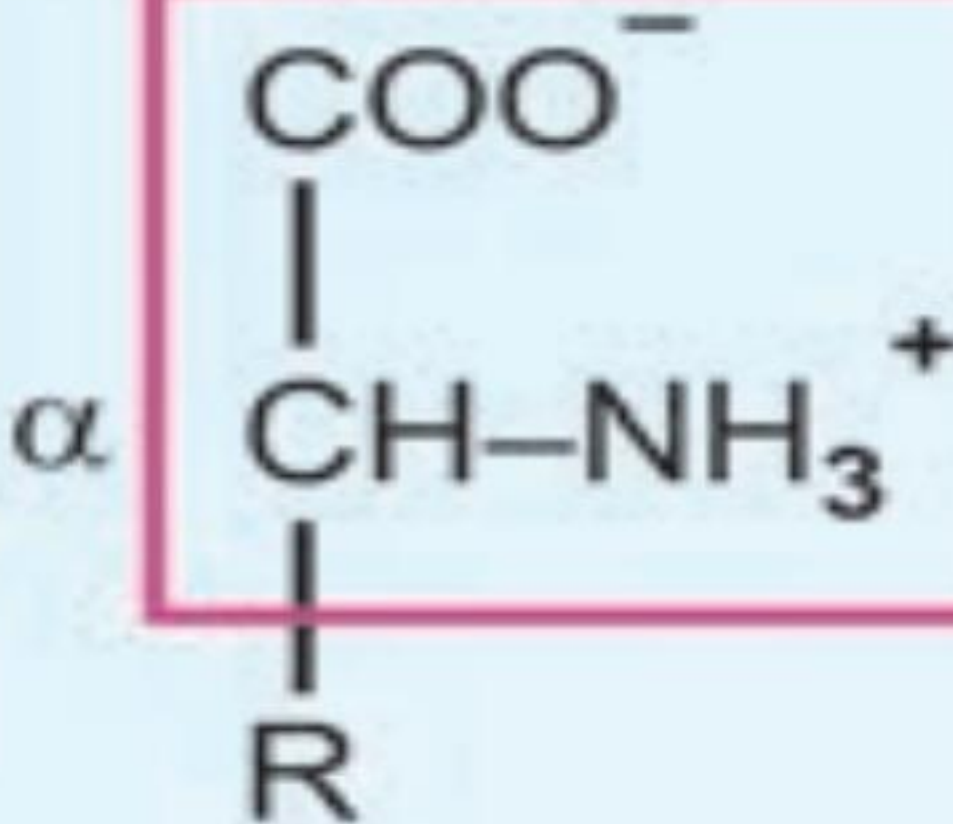




Amino Acids

By: Alie M.

Outline

- Definition and basic structure of amino acids
- Classification of amino acids based on structure
- Based on side chain character
- Based on metabolic fate
- Based on nutritional requirements
- Iso electric point
- Reactions due to carboxyl group , amino group , SH group ,
- Optical activity
- Peptide bond formation

- **Amino acids:** Are the basic building blocks of proteins.
- Proteins are of paramount importance for biological systems. All the major structural and functional aspects of the body are carried out by protein molecules. All proteins are polymers of amino acids. Proteins are composed of a number of amino acids linked by **peptide bonds**. Although about 300 amino acids occur in nature, only 20 of them are seen in human body. Most of the amino acids (except **proline**) are **alpha amino acids**, which means that the amino group is attached to the same carbon atom to which the carboxyl group is attached.

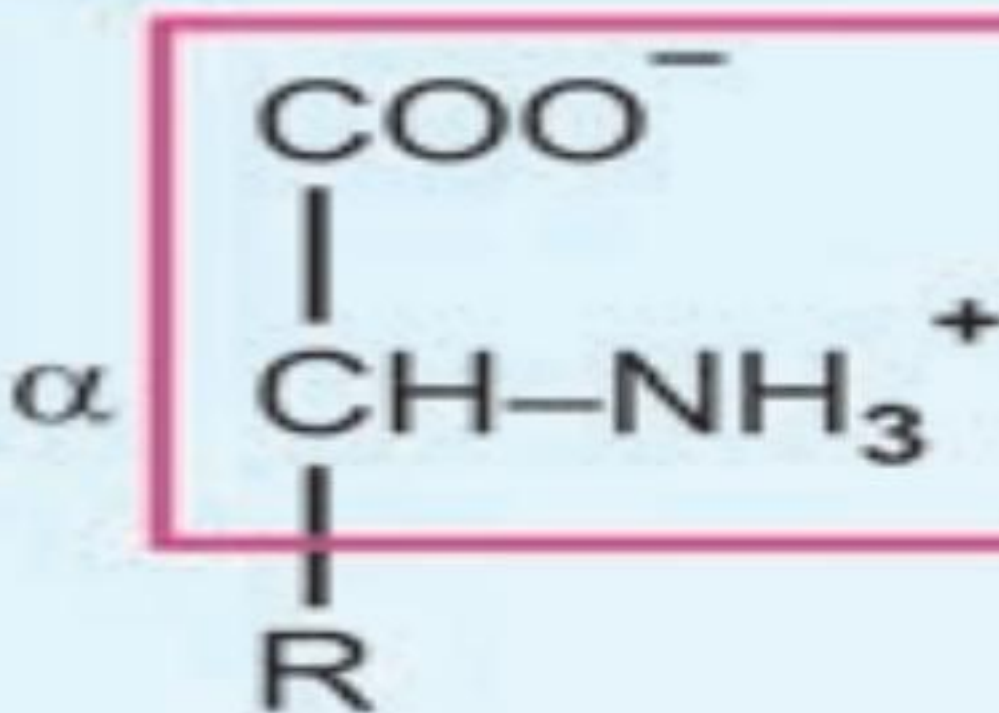


Fig.1. Basic structure of amino acids

CLASSIFICATION OF AMINO ACIDS

1. Based on Structure

1-A. Aliphatic amino acids

a. Mono amino mono carboxylic acids:

- Simple amino acids: Glycine, Alanine
- Branched chain amino acids: Valine, Leucine, Isoleucine
- Hydroxy amino acids: Serine, Threonine
- Sulphur containing amino acids: Cysteine, Methionine
- Amino acids with amide group: Asparagine, Glutamine

b. Mono amino dicarboxylic acids: Aspartic acid, Glutamic acid

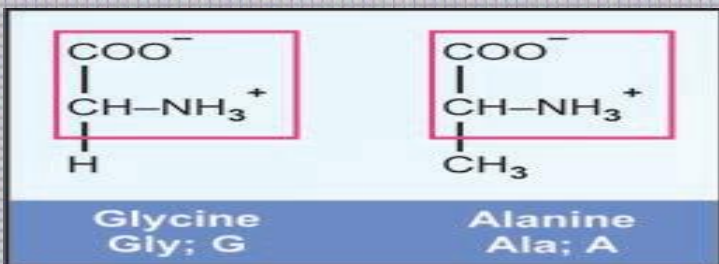


Fig 2. Simple amino acids

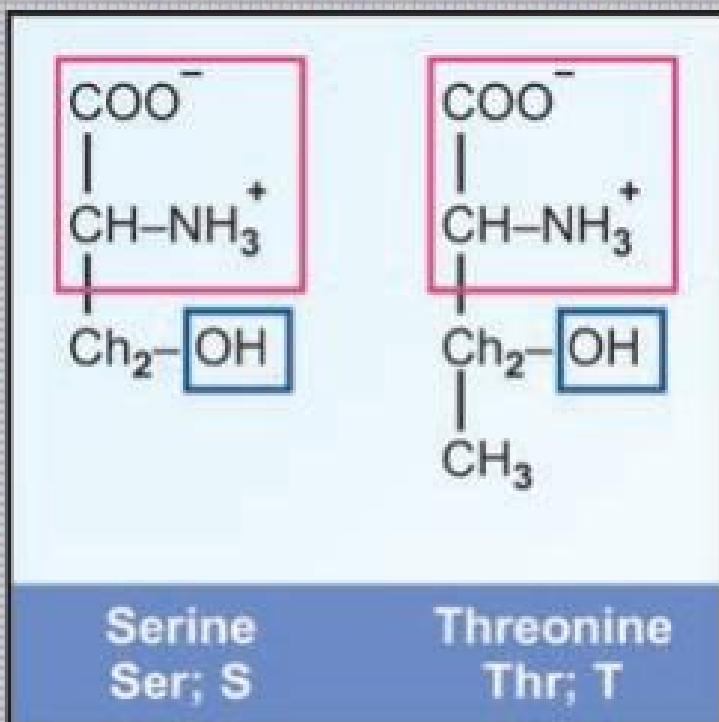


Fig .4. Hydroxy amino acids

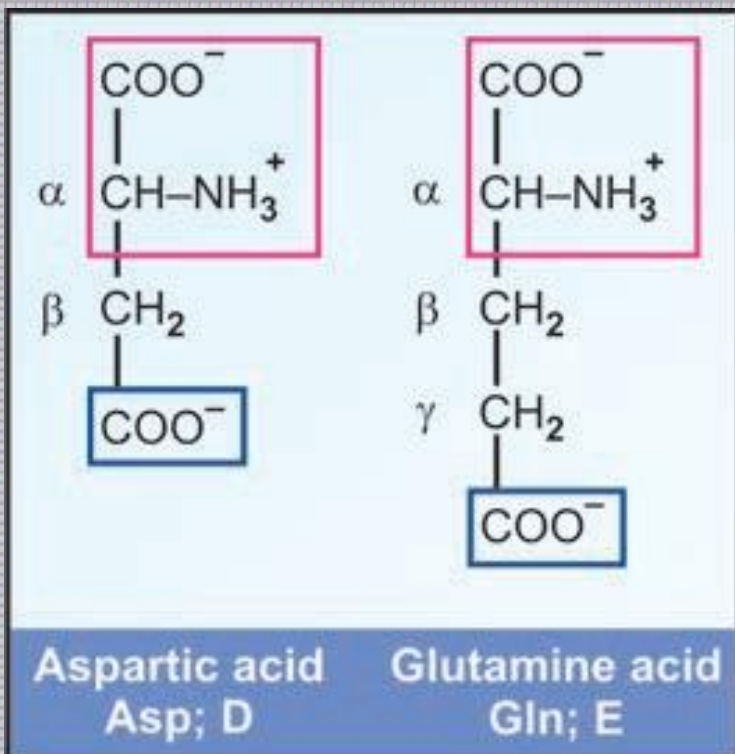


Fig. 6. Dicarboxylic amino acids

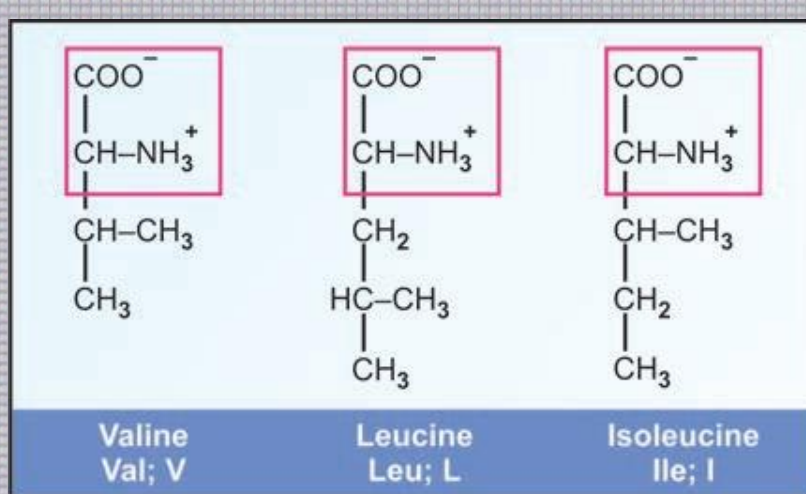


Fig .3. Branched chain amino acids

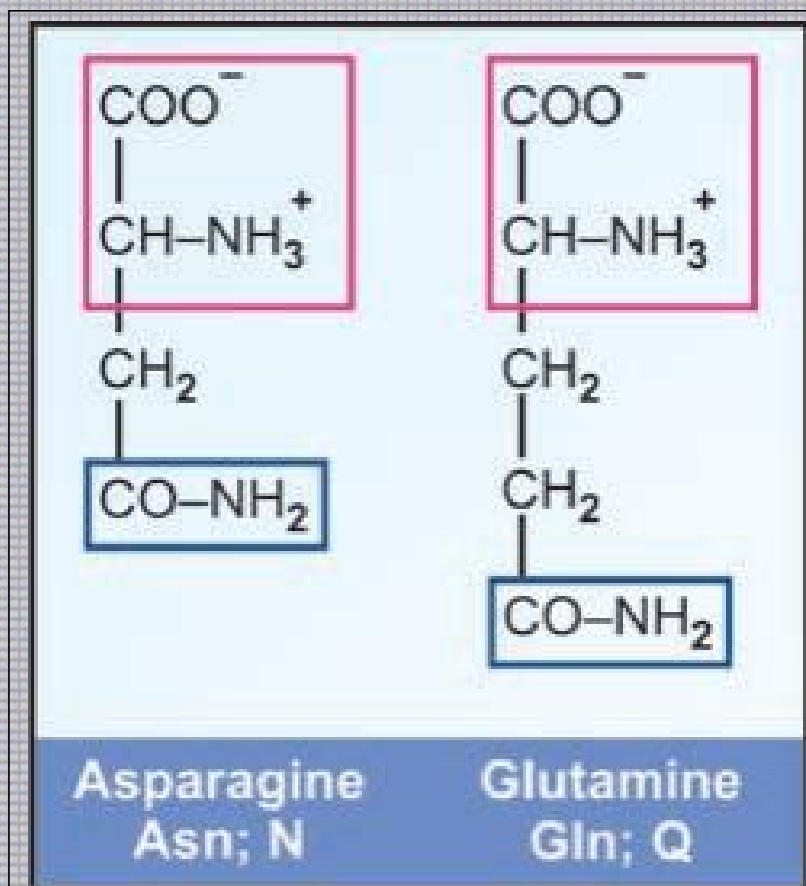


Fig. 5. Amino acids with amide groups

c. Di basic mono carboxylic acids: Lysine, Arginine

1-B. Aromatic amino acids: Phenylalanine, Tyrosine

1-C. Heterocyclic amino acids: Tryptophan, Histidine

1-D. Imino acid: Proline

1-E. Derived amino acids:

1-E-i. Derived amino acids found in proteins: After the synthesis of proteins, some of the amino acids are modified, e.g. hydroxy proline and hydroxy lysine are important components of collagen. Gamma carboxylation of glutamic acid residues of proteins is important for clotting process. In ribosomal proteins and in histones, amino acids are extensively methylated and acetylated.

1-E-ii. Derived amino acids not seen in proteins

(Non-protein amino acids):

Some derived amino acids are seen free in cells, e.g. Ornithine, Citrulline, Homocysteine. These are produced during the metabolism of amino acids. Thyroxine may be considered as derived from tyrosine.

1-E-iii. Non-alpha amino acids:

Gamma amino butyric acid (GABA) is derived from glutamic acid. Beta alanine, where amino group is in beta position, is a constituent of pantothenic acid (vitamin) and co-enzyme A.

Special Groups in Amino Acids

In the figures, special groups are shaded. Arginine contains **guanidinium** group; Phenyl alanine (**benzene**); Tyrosine (**phenol**); Tryptophan (**Indole**); Histidine (**imidazole**); and Proline (**pyrrolidine**). Proline has a secondary amino group, and hence it is an **imino acid**.

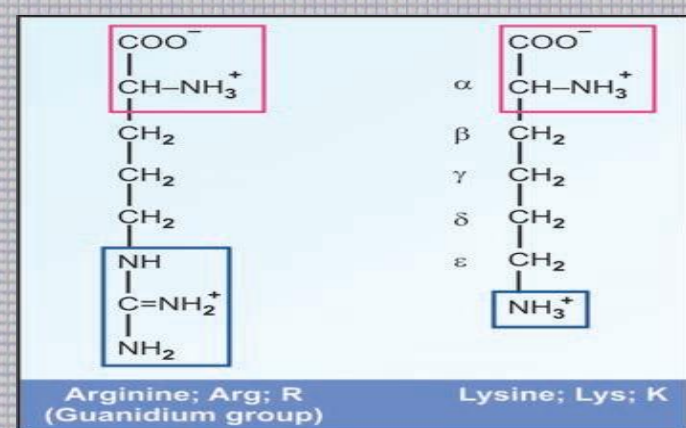


Fig.8.Di basic amino acids

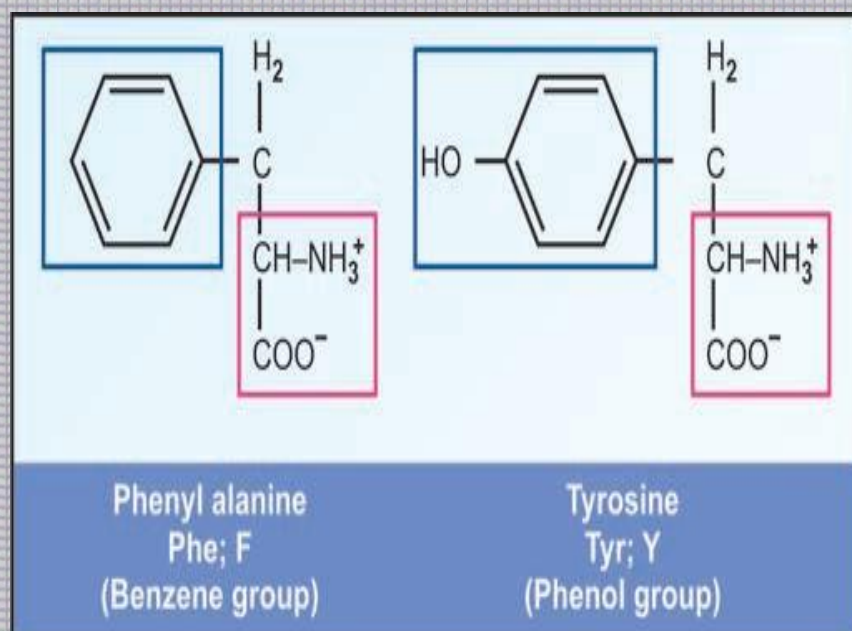


Fig.7. Aromatic amino acids

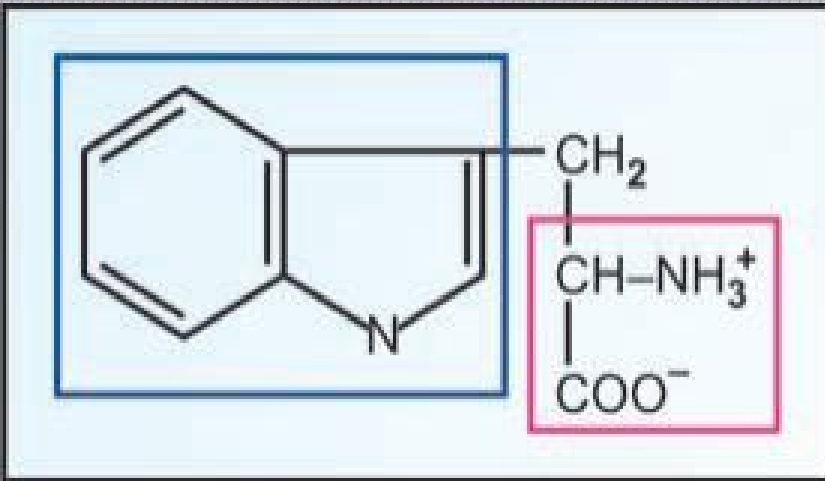


Fig. 9. Tryptophan (Trp) (W) with indole group

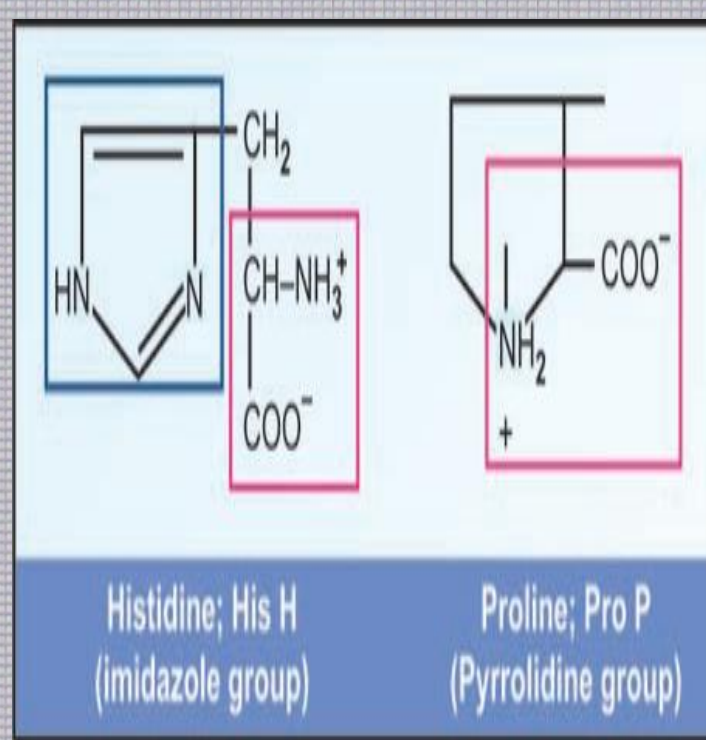


Fig. 10. Histidine and proline

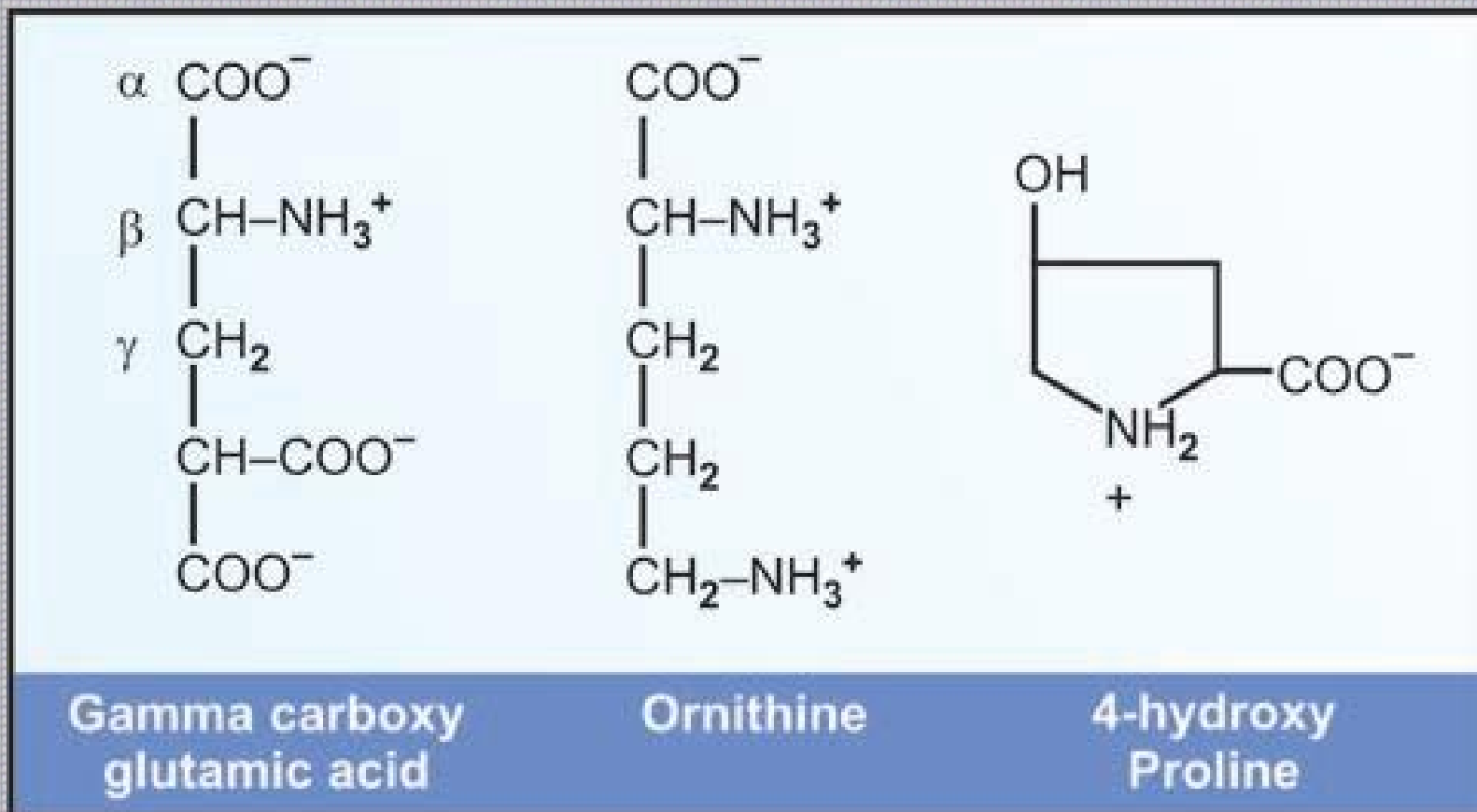


Fig. 11. Some derived amino acids

2. CLASSIFICATION BASED ON SIDE CHAIN

2-A. Amino acids having nonpolar side chains: These include Alanine, Valine, Leucine, Isoleucine, Methionine, Proline, Phenylalanine and Tryptophan. These groups are **hydrophobic** (water repellant) and lipophilic. Therefore, the parts of proteins made up of these amino acids will be hydrophobic in nature.

2-B. Amino acids having uncharged or non ionic polar side chains: Glycine, Serine, Threonine, Cysteine, Tyrosine, Glutamine and Asparagine belong to this group. These amino acids are **hydrophilic** in nature. (Tyrosine and Cysteine may show hydrophobic character when present in the interior of the protein).

2-C. Amino acids having charged or ionic polar side chains (hydrophilic):

C-a. Acidic amino acids: They have a negative charge on the R group:

Aspartic acid and Glutamic acid (Tyrosine is mildly acidic).

C-b. Basic amino acids: They have a positive charge on the R group: Lysine, Arginine and Histidine.

3. CLASSIFICATION BASED ON METABOLISM

3-A. Purely Ketogenic : Leucine is purely ketogenic because it is converted to ketone bodies

3-B. Ketogenic and Glucogenic : Lysine, Isoleucine, Phenylalanine, Tyrosine and Tryptophan are partially ketogenic and partially glucogenic. However in humans lysine is predominantly ketogenic. During metabolism, part of the carbon skeleton of these amino acids will enter the ketogenic pathway and the other part to glucogenic pathway .

3-C. Purely Glucogenic : All the remaining 14 amino acids are purely glucogenic as they enter only into the glucogenic pathway .

4. CLASSIFICATION BASED ON NUTRITIONAL REQUIREMENTS

4-A. Essential or Indispensable ; The amino acids may further be classified according to their essentiality for growth. Thus **Isoleucine, Leucine, Threonine, Lysine, Methionine, Phenylalanine, Tryptophan, and Valine** are essential amino acids. Their carbon skeleton cannot be synthesized by human beings and so preformed amino acids are to be taken in food for normal growth.

4-B. Partially essential or Semi-essential : Histidine and arginine are semi-indispensable amino acids. Growing children require them in food. But they are not essential for the adult individual.

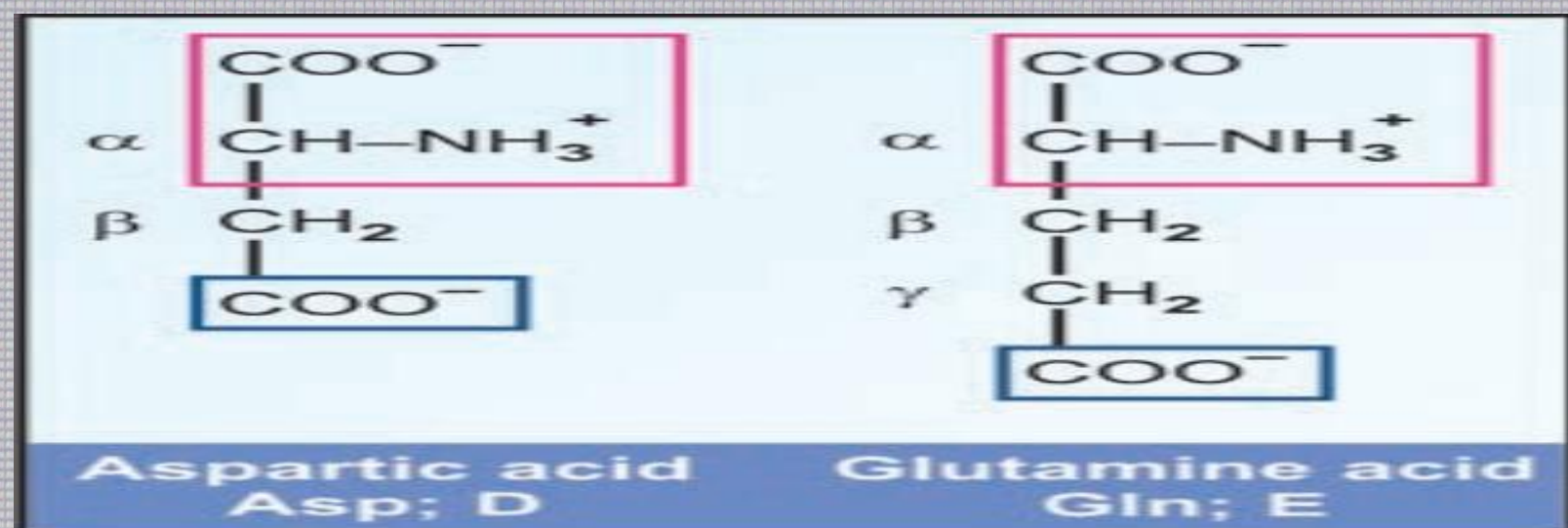
4-C. Non-essential or Dispensable : The remaining 10 amino acids are non-essential because their carbon skeleton can be synthesized by the body. However, they are also required for normal protein synthesis. All body proteins do contain all the non-essential amino acids.

NB: 21st century have witnessed the addition of **selenocysteine** and **pyrolysine** as a newly invented amino acids present in human proteins.

Table 3.1. Common amino acids

Name of amino acid	Special group present	3-letter abbreviation	1-letter abbreviation	Molecular weight (D)
Glycine		Gly	G	77
Alanine		Ala	A	89
Valine		Val	V	117
Leucine		Leu	L	131
Isoleucine		Ile	I	131
Serine	Hydroxyl	Ser	S	105
Threonine	Hydroxyl	Thr	T	119
Cysteine	Sulfhydryl	Cys	C	121
Methionine	Thioether	Met	M	149
Asparagine	Amide	Asn	N	132
Glutamine	Amide	Gln	Q	146
Aspartic acid	Beta-carboxyl	Asp	D	133
Glutamic acid	γ -carboxyl	Glu	E	147
Lysine	ϵ -amino	Lys	K	146
Arginine	Guanidinium	Arg	R	174
Phenylalanine	Benzene	Phe	F	165
Tyrosine	Phenol	Tyr	Y	181
Tryptophan	Indole	Trp	W	204
Histidine	Imidazole	His	H	155
Proline (imino acid)	Pyrrolidine	Pro	P	115

Naming (Numbering) of Carbon Atoms: Carbon atoms in amino acids in sequence are named with letters of Greek alphabets, starting from the carbon atom to which carboxyl group is attached. Example: Naming of E and D



PROPERTIES OF AMINO ACIDS : Glycine, alanine, valine, serine, tryptophan, histidine and proline are sweet in taste; leucine is tasteless; while isoleucine and arginine are bitter. Sodium glutamate is a flavoring agent. Aspartame, an artificial sweetener contains aspartic acid and phenyl alanine. All amino acids have high melting points (more than 200°C). All amino acids are soluble in water and alcohol (polar solvents); but insoluble in nonpolar solvents (benzene).

A. Ampholyte and Iso-electric Point : Amino acids can exist as **ampholytes** or **zwitterions** (German word "zwitter" = hybrid). Ampholyte is a molecule that can act both as an acid and a base, depending on the environment. Ampholytes contain both acidic and basic functional groups, allowing them to either donate or accept protons (H^+ ions). This dual behavior is particularly evident in aqueous solutions, where the pH determines whether the molecule will act as an acid or a base. **Ampholytes** are often used in isoelectric focusing, a technique used to separate molecules like proteins based on their isoelectric point (pI), where the molecule carries no net charge. Iso electric point is the pH at which a particular molecule, typically a protein or amino acid, carries no net electrical charge. At the isoelectric point, the molecule is electrically neutral because the number of positive and negative charges on the molecule balance each other out. For **ampholytes** like amino acids or proteins, which contain both acidic and basic functional groups, the pI is important because it defines the pH at which the molecule exists predominantly in its zwitterionic form. In this form, the molecule has both positive and negative charges, but the total charge is zero.

At the isoelectric point:

- The molecule does not move in an electric field because it has no net charge.
- The solubility of the molecule in water can be lower at the pI, making it prone to precipitation in some cases.

In **proteins**, the pI is determined by the specific sequence and number of ionizable side chains. **Isoelectric focusing**, a common laboratory technique, exploits this property to separate proteins based on their pI by applying an electric field to a pH gradient, allowing proteins to migrate to the pH that matches their isoelectric point.

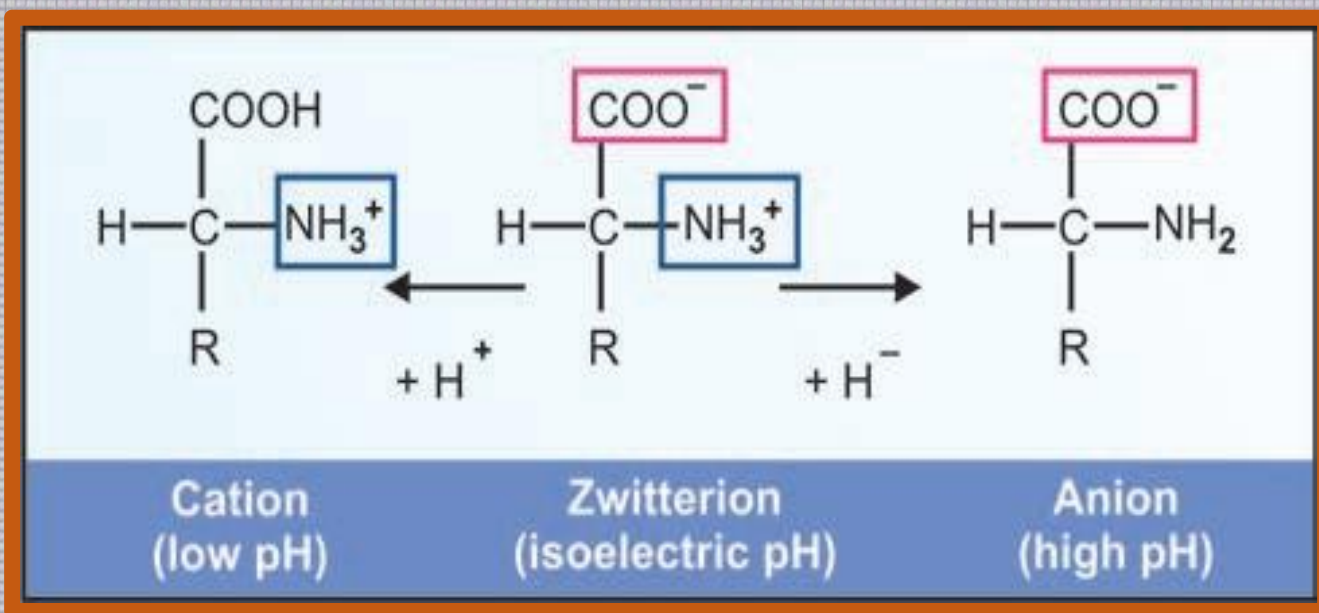


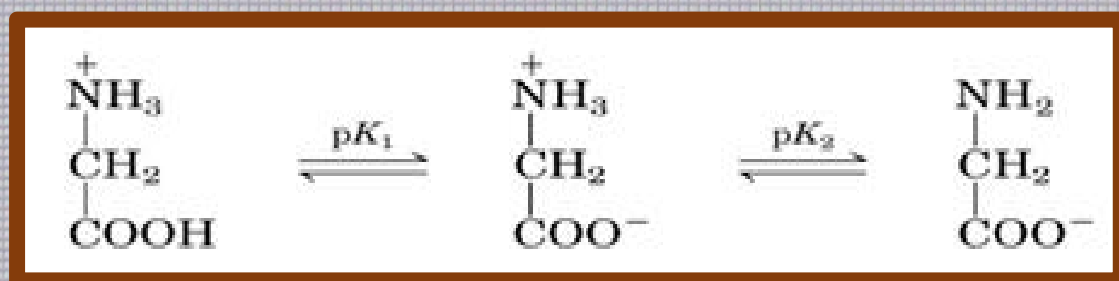
Fig. 12. Ionic forms of amino acids

Amino Acid titration curve: Refers to the process of gradually adding a titrant (typically a strong acid or base) to an amino acid solution and monitoring how the pH changes. This process helps determine the ionization states of the amino acid's functional groups (such as the amino group, carboxyl group, and any ionizable side chains) at different pH levels. The titration curve generated from this process allows you to visualize the points at which the amino acid donates or accepts protons, and to estimate important properties like pKa values and the **isoelectric point (pI)**. Amino acids have at least two ionizable groups: **Amino group (-NH₃⁺)**, which can lose a proton and become neutral. **Carboxyl group (-COOH)**, which can lose a proton and become negatively charged (-COO⁻). Some amino acids may have additional ionizable side chains (like the -R group in lysine, glutamic acid, etc.). **pKa Values:** During titration, the pH at which each functional group is half ionized (50% protonated, 50% deprotonated) is called its **pKa**. Amino acids usually have: A **pKa₁** for the carboxyl group (typically around pH 2-3). A **pKa₂** for the amino group (typically around pH 9-10). If the amino acid has an ionizable side chain (like histidine, lysine, etc.), there will be a third pKa for that group.

Titration Process:

- A **strong acid** (like HCl) is titrated into a solution of an amino acid to decrease the pH, or a **strong base** (like NaOH) is added to raise the pH.
- As the pH changes, the amino acid's functional groups become protonated or deprotonated depending on the pH.
- By plotting the pH against the amount of titrant added, you obtain a **titration curve**. The **flat regions** on the curve correspond to buffer zones where the amino acid resists changes in pH, indicating that a functional group is being titrated (a proton is being gained or lost).

The Titration curve of glycine



1. At a very low pH (acidic) both groups are fully protonated where the solution predominantly contains: Net charge of the solution is +1



2. When the pH is raised, the -COOH group start to be deprotonated and the proportion will be:



3. $\text{pH} = \text{pK}_a$, where it will act as a buffer and the solution will contain an equal amount of :



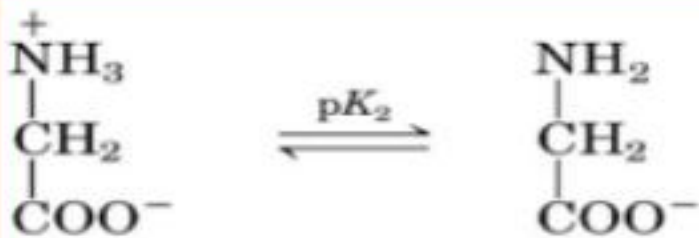
4-Further increase in pH, the solution will predominantly contains zwitter ion and the pH at this point is equal to pI. Net charge of the solution is zero.



5-As the pH increases, the second group NH_3^+ will be deprotonated;

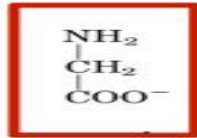


6- After that, $\text{pH} = \text{pK}_a$ where it will as a buffer and the solution will contain an equal amount of:

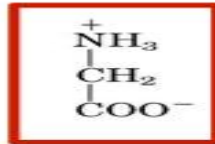


7. Further increase in pH the solution predominantly contains:

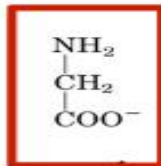
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8. The NH_3^+ group will dissociate and at the same time full dissociation of glycine takes place



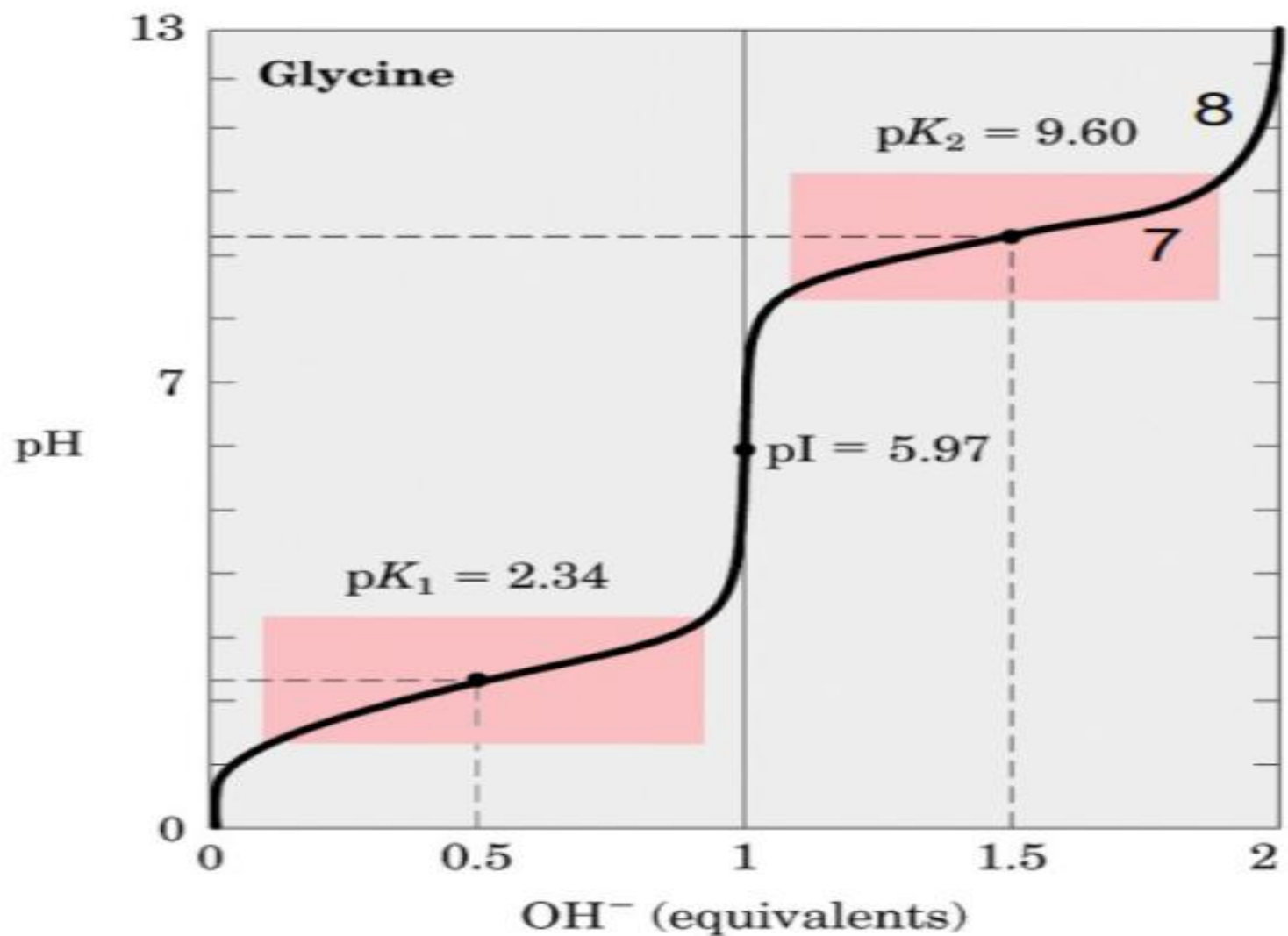


Fig .13.Titration curve of glycine

pI Formula for Neutral Amino Acids (No Ionizable Side Chain):

$$\text{pI} = \frac{\text{pKa}_1(\text{Carboxyl group}) + \text{pKa}_2(\text{Amino group})}{2}$$

2

pI Formula for Acidic Amino Acids (Aspartic Acid, Glutamic Acid): acidic amino acids like aspartic acid and glutamic acid have an additional carboxyl group in their side chain, which is ionizable. For acidic amino acids, the pI is the average of the pKa values of the two carboxyl groups: the side chain carboxyl group and the α -carboxyl group.

$$\text{pI} = \frac{\text{pKa}_1(\alpha\text{ carboxyl group}) + \text{pKa}_2(\text{R carboxyl group})}{2}$$

2

pI Formula for Basic Amino Acids (Lysine, Arginine, Histidine):

Basic amino acids like **lysine**, **arginine**, and **histidine** have an ionizable amino group in their side chain. For basic amino acids, the pI is the average of the pKa values of the **amino group** and the **R group** (side chain).

$$pI = \frac{pK_a 1(\text{Alpha amino group}) + pK_a 2(\text{R amino group})}{2}$$

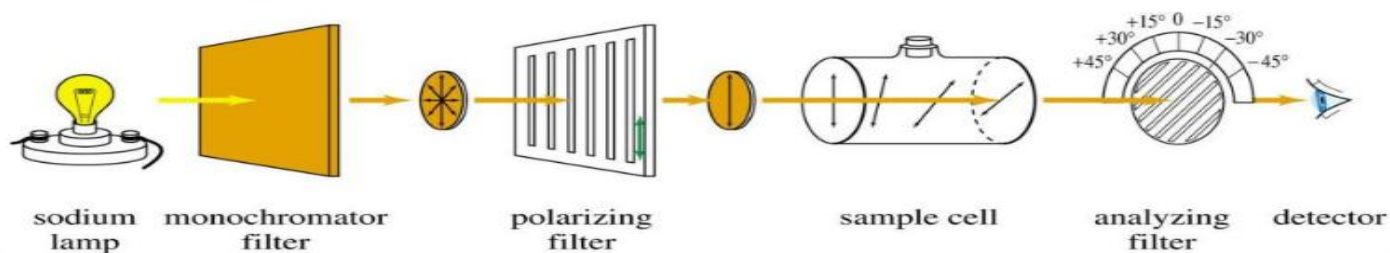
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B. **Optical Activity** : Refers to the ability of certain substances to rotate the plane of polarized light as it passes through them. This property is a characteristic of **chiral molecules**, which have structures that are not superimposable on their mirror images (like left and right hands). The rotation occurs because the substance interacts with the light's electric field, causing the light's plane of polarization to rotate either to the right (clockwise) or to the left (counterclockwise). Amino acids having an asymmetric carbon atom exhibit optical activity. **Asymmetry** arises when 4 different groups are attached to the same carbon atom. Glycine is the simplest amino acid and has no asymmetric carbon atom and therefore shows no optical activity. All others are optically active. The mirror image forms produced with reference to the alpha carbon atom, are called **D and L isomers**. The L-amino acids occur in nature and are therefore called **natural amino acids**. D-amino acids are seen in small amounts in microorganisms and as constituents of certain antibiotics such as Gramicidin-S, Polymyxin, Actinomycin-D and Valinomycin, as well as bacterial cell wall peptidoglycans. Isoleucine and threonine have 2 optically active centers and therefore each has 4 diastereoisomers.

Optical Activity

- When plane polarized light passes through a solution containing a **single** chiral compound, the chiral compound causes the plane of vibration to rotate.

- **Polarimeter**



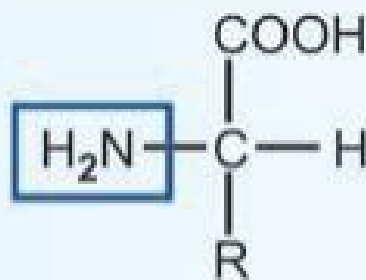
Steps to Assign D and L Configuration for Amino Acids:

Draw the Amino Acid in Fischer Projection:

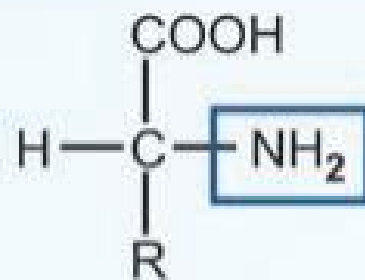
- A Fischer projection is a way of representing the molecule where the chiral carbon is shown as a cross, with four substituents attached.
- The **carboxyl group (-COOH)** is placed **on the top**.
- The **R group** (side chain of the amino acid) is placed **at the bottom**.
- The **amino group (-NH₂)** and **hydrogen (-H)** are placed on the **horizontal line**, to the left and right of the central carbon, respectively.

Identify the Position of the Amino Group (-NH₂):

- If the **amino group (-NH₂)** is on the **left** of the chiral carbon in the Fischer projection, the amino acid is in the **L-configuration**.
- If the **amino group (-NH₂)** is on the **right** of the chiral carbon in the Fischer projection, the amino acid is in the **D-configuration**.



L-amino acid

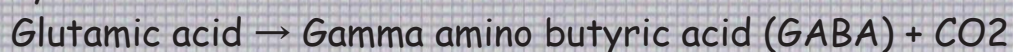
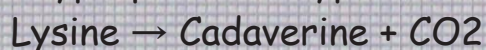
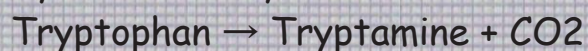
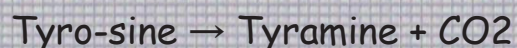
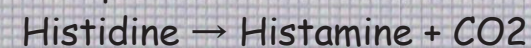


D-amino acid

GENERAL REACTIONS OF AMINO ACIDS

A. Due to Carboxyl Group

1. Decarboxylation: The amino acids will undergo alpha decarboxylation to form the corresponding amine. Thus some important amines are produced from amino acids. For example,

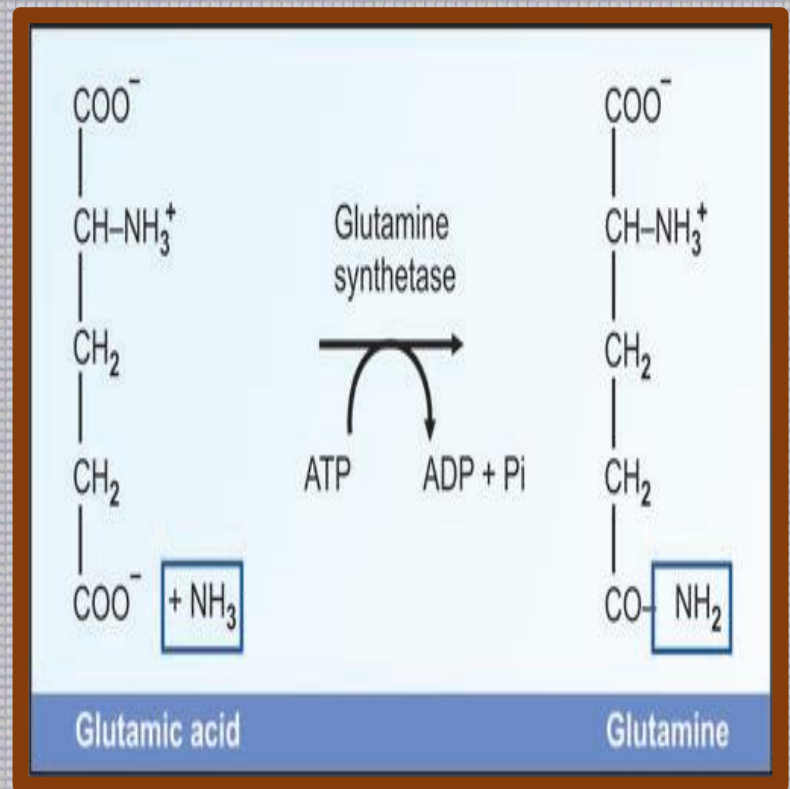
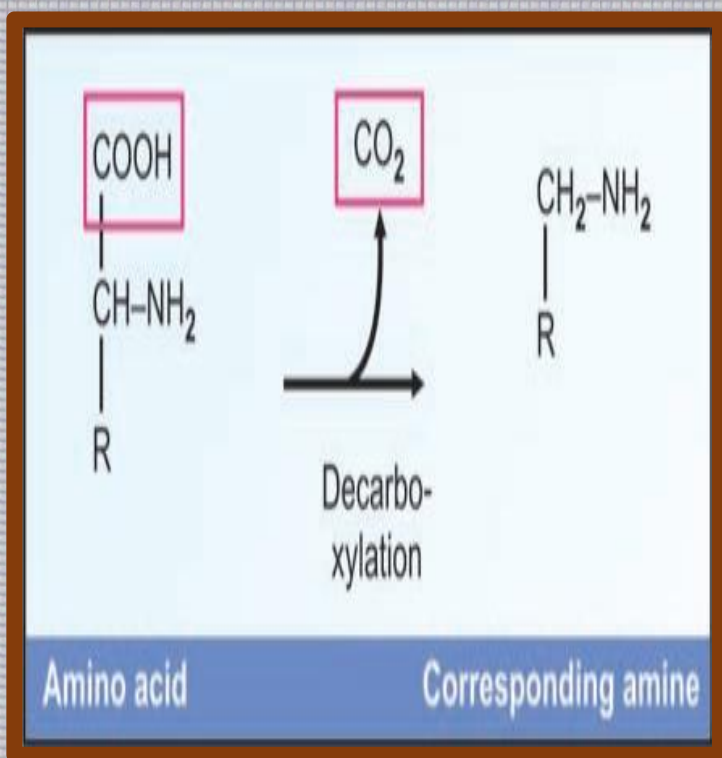


2. Amide Formation: The $-COOH$ group of dicarboxylic amino acids (other than alpha carboxyl) can combine with ammonia to form the corresponding amide. For example,

Aspartic acid + $NH_3 \rightarrow$ Asparagine

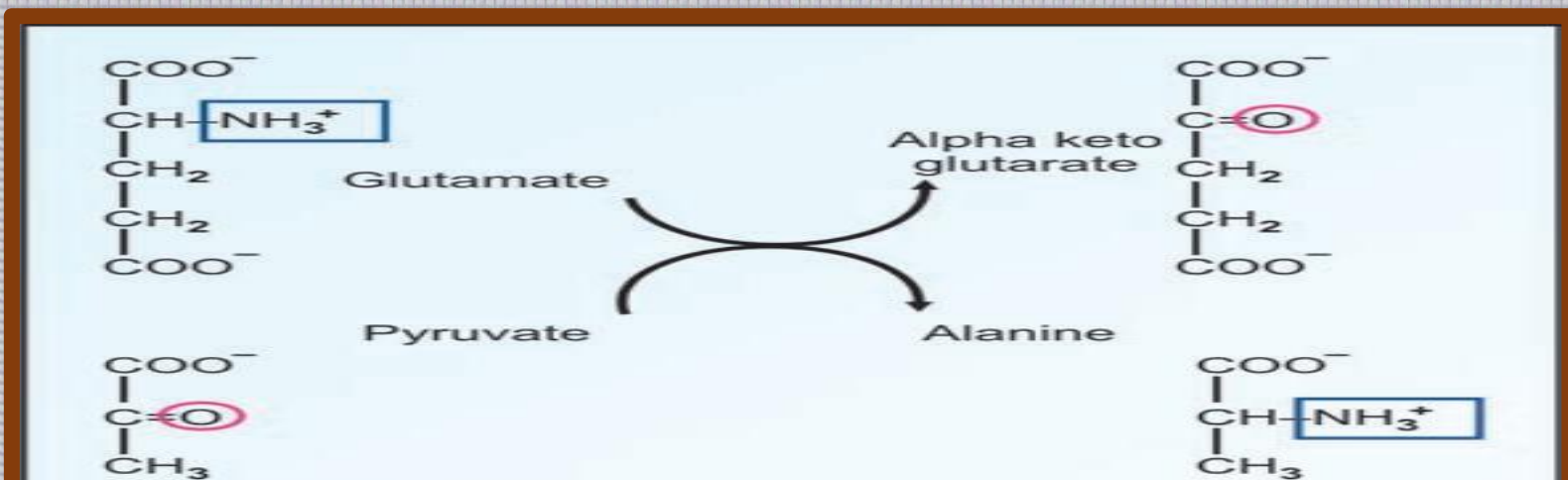
Glutamic acid + $NH_3 \rightarrow$ Glutamine

These amides are also components of protein structure. The amide group of glutamine serves as the source of nitrogen for nucleic acid synthesis.



B. Reactions Due to Amino Group

3. Transamination: The alpha amino group of amino acid can be transferred to alpha keto acid to form the corresponding new amino acid and alpha keto acid. This is an important reaction in the body for the inter-conversion of amino acids and for **synthesis of non-essential amino acids**.



4. Oxidative Deamination: The alpha amino group is removed from the amino acid to form the corresponding keto acid and ammonia. In the body, **Glutamic acid** is the most common amino acid to undergo oxidative deamination.

5. Formation of carbamino compound: Carbon dioxide adds to the alpha amino group of amino acids to form carbamino compounds. The reaction occurs at alkaline pH and serves as a mechanism for the transport of carbon dioxide from tissues to the lungs by hemoglobin. $\text{Hb-NH}_2 + \text{CO}_2 \rightarrow \text{Hb-NH-COOH}$ (Carbamino-Hb)

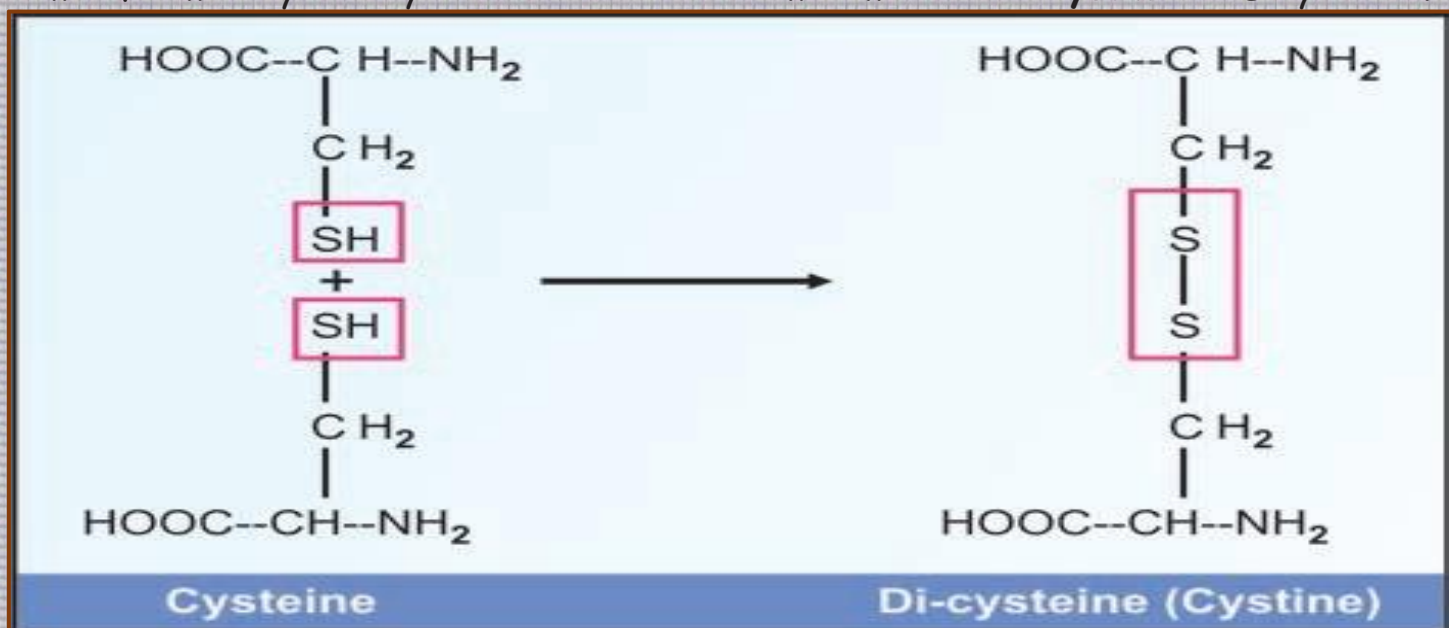
C. Reactions Due to Side Chains

6. Transmethylation: The methyl group of Methionine, after activation, may be transferred to an acceptor which becomes methylated. $\text{Methionine} + \text{Acceptor} \rightarrow \text{Methylated Acceptor} + \text{Homocysteine}$

7. Ester formation by the OH group: The hydroxy amino acids can form esters with phosphoric acid. In this manner the **Serine and Threonine** residues of proteins are involved in the formation of phosphoproteins. Similarly these hydroxyl groups can form O-glycosidic bonds with carbohydrate residues to form glycoproteins.

8. Reaction of the amide group: The amide groups of **Glutamine and Asparagine** can form N glycosidic bonds with carbohydrate residues to form glycoproteins.

9. Reactions of SH group: Cysteine has a sulfhydryl (SH) group and it can form a disulphide (S-S) bond with another cysteine residue. The two cysteine residues can connect two polypeptide chains by the formation of **interchain disulphide bonds**. The dimer formed by two cysteine residues is sometimes called **Cystine** or Dicysteine.

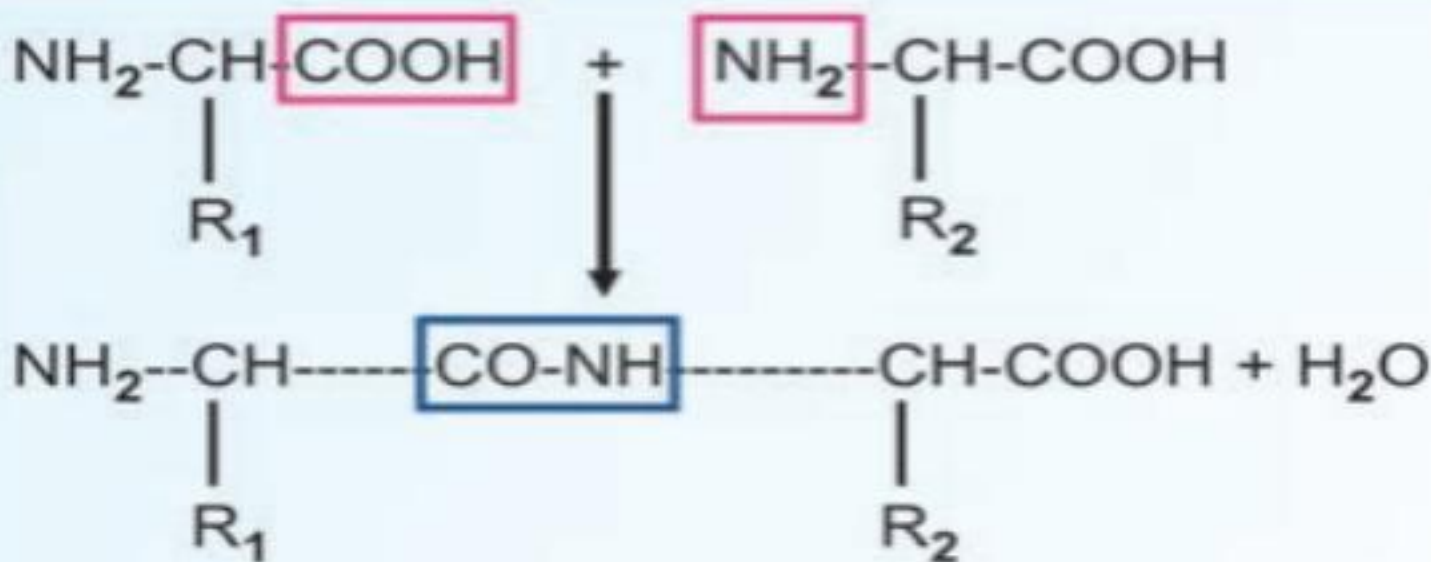


Amino Acid Derivatives of Importance

- i. **Gamma amino butyric acid** (GABA, a derivative of glutamic acid) and dopamine (derived from tyrosine) are neurotransmitters. **Gabapentin** (an analog of GABA) can pass blood brain barrier and can form GABA in brain. Gabapentin is clinically used to relieve pain.
- ii. **Histamine** (synthesized from histidine) is the mediator of allergic reactions.
- iii. **Thyroxine** (from tyrosine) is an important thyroid hormone.
- iv. **Cycloserine**, a derivative of serine is an antituberculous drug. **Azaserine** inhibits reactions where amide groups are added, and so acts as an anticancer drug.
- v. **Histidine** residues are important in the buffering activity of proteins.
- vi. **Ornithine** and citrulline are derivatives of arginine, and are essential for urea synthesis.

PEPTIDE BOND FORMATION

Alpha carboxyl group of one amino acid reacts with **alpha amino group** of another amino acid to form a peptide bond or CO-NH bridge. Proteins are made by polymerization of amino acids through peptide bonds.



Color Reactions of Amino Acids and Proteins

1. **Ninhydrin Reaction** (Ruhemann, 1910) heat Amino acid + 2 molecules of ninhydrin -----
----- Aldehyde with one carbon atom less + CO_2 + purple complex. All amino acids when heated with ninhydrin can form complexes; pink, purple or blue in color. The color complex is called **Ruhemann's purple**. Proline and hydroxy proline will give yellow color with ninhydrin. Amino acids with amide group (glutamine, asparagine) produce a brown color. The ninhydrin reaction may be adopted for qualitative as well as quantitative estimation of amino acids. It is often used for detection of amino acids in chromatography. Proteins do not give a true color reaction; but N-terminal end amino group of protein will also react with ninhydrin, to produce a blue color.

2. Biuret Reaction: Cupric ions in an alkaline medium form a violet color with peptide bond nitrogen (Schiff, 1896). This needs a minimum of two **peptide bonds**, and so *individual amino acids and di-peptides will not answer this test*. This reaction can be used for quantitative estimation also. The name is derived from the compound biuret ($\text{NH}_2\text{—CO—NH—CO—NH}_2$), a condensation product of two urea molecules, which also gives a positive color test. Magnesium and ammonium sulphates interfere with this reaction.

3. Xanthoproteic test: The ring systems in **phenyl alanine, tyrosine and tryptophan** undergo nitration on treatment with concentrated nitric acid when heated (Salkowski, 1888). The end product is yellow in color which is intensified in strong alkaline medium. This reaction causes the yellow stain in skin by nitric acid.

4. Millon's Test: The phenol group of **phenylalanine and tyrosine** containing proteins, when heated with mercuric sulphate in sulphuric acid and sodium nitrite (or, mercurous and mercuric nitrates in nitric acid) form red colored mercury phenolate (Millon, 1849). Chloride interferes with this reaction and so it is not suitable to test for tyrosine in urine samples. Both xanthoproteic and Millon's tests are negative for tapioca (casava) which is deficient in phenylalanine and tyrosine.

5. Aldehyde tests for tryptophan: In the **Hopkins-Cole test**, tryptophan containing protein is mixed with glyoxylic acid, and the mixture is layered over concentrated sulphuric acid. A violet ring at the interface of liquids shows the presence of the indole ring. Formaldehyde and mercuric sulphate is used similarly in **Acree-Rosenheim reaction** to get a violet color. Para-dimethyl-amino-benzaldehyde and strong hydrochloric acid give dark blue color (Ehrlich's reaction). Gelatin with limited tryptophan content will not answer these tests.

5. Sakaguchi's test for arginine: Free arginine or arginyl residues in proteins react with alpha-naphthol and alkaline hypobromite to give bright red color. This is due to the guanidinium group.

7. Sulphur test for cysteine: When cysteine or cysteinecontaining proteins are boiled with strong alkali, organic sulphur splits and forms sodium sulphide, which on addition of lead acetate produces lead sulphide as a black precipitate. Methionine does not answer this test because sulphur in methionine is in the thio-ether linkage which is difficult to break. Albumin and keratin will answer sulphur test positively; but casein will give a negative test.

8. Nitroprusside reaction for SH groups: Proteins with free sulfhydryl groups give a reddish color with sodium nitroprusside, in ammoniacal solution. Many proteins give a negative, reaction in the native state, but when denatured, reaction will be positive, showing the emergence of free SH groups.

9. Pauly's test for histidine or tyrosine: Diazo benzene sulfonic acid reacts with imidazole group of Histidine to form a cherry-red colored diazotised product under alkaline conditions. The same reagent will give an orange red colored product with phenol group of Tyrosine.

Table 3.. Color reactions of amino acids

Reaction	Answered by specific group
1. Ninhydrin	Alpha amino group
2. Biuret reaction	Peptide bonds
3. Xanthoproteic test	Benzene ring (Phe, Tyr, Trp)
4. Millon's test	Phenol (Tyrosine)
5. Aldehyde test	Indole (Tryptophan)
6. Sakaguchi's test	Guanidinium (Arginine)
7. Sulphur test	Sulfhydryl (Cysteine)
8. Nitroprusside test	Sulfhydryl (Cysteine)
9. Pauly's test	Imidazole (Histidine)

Stay positive, like arginine!
I owe you one!.

